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????? batch ????small-batch panels ? fold-matched full Top2000 panels ???? prior-union genes???
emerged prior genes ?? SFRP1?MIA?PGR?GPR160?FOXA1 ? MLPH????? proliferation-associated
PAM50 genes??

????? baseline ???????

Top30 XGBoost?logistic regression ? elastic net ??????????????????????????????METABRIC
????????????? primary-tumour ???full/consensus primary metrics ??????? 0.756 ?
0.993????????????????????????????? small-batch consensus ?????????? full-space panel?

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????????????????????????????? resampling ???? split-seed instability ?????????????????? full-space
masking??
??????? small-batch random-subspace XGBoost ??????? CV5 ?????????? Top100 ?????????????? prior
genes?Poster ???????? leakage control?stability auditing ? candidate prioritisation?????????????????
biomarker?

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Figure 1. ?????? random-subspace screening workflow??? ranking?panel export ? validation ???? fold
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Figure 2. Small-batch screening ?? Top100 reproducibility????????? CV5 performance?a, full Top100
??? small-batch/consensus Top100 Jaccard?b, ?? small-batch Top2000 ?? full Top2000 ? primary-
metric delta?c, stable prior-gene gains ? stable emerged Top100 prior genes?

Figure 3. Baselines ? METABRIC ??????a, XGBoost?logistic regression ? elastic net ? Top30
?????????????b, METABRIC ?? primary metrics ? full Top2000 ? small-batch consensus ?????